TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

PROJECT SYNOPSIS

Student Details

Institute/University:

_____

Programme / Batch:

_____

Student Name: _____

PRN: _____

1. Case Study ID and Project Title / Topic

Case Study ID (CS1 to CS5):

_____

Project Title / Topic:

_____

2. Objectives

Define the specific objectives and focus areas of the allocated AIML case study.

Objectives

- _____
- _____
- _____
- _____

3. Scope

Clearly state what will be implemented in the project and what will not be covered. Respect mandatory controls such as citations, human review, and approved or synthetic data.

In Scope – To Be Implemented

- _____
- _____
- _____
- _____

Out of Scope – Will Not Be Implemented

- _____
- _____
- _____

4. Proposed Approach

Pre-trained model (name, version, local/private hosting)

Embedding model

Vector store (FAISS / ChromaDB)

RAG pipeline and/or AI-agent design – brief description

Service layer and UI (e.g., FastAPI / Flask / Streamlit)

Tools / deterministic rules, if applicable

Proposed System / Architecture

[Attach system architecture / workflow diagram here]

5. Input Data / Knowledge Base

Use approved or synthetic data only.

Data / Knowledge Sources

Approximate Size and Format

Preprocessing and EDA Planned

6. Expected Outcomes and Evaluation Plan

Expected Outputs – aligned to the case study

Evaluation Method – test question set / ground truth / metrics / validation approach

_____

_____

Responsible AI Measures – grounding and citations / privacy / security / human oversight

_____

_____

MLOps / Deployment Elements Planned – versioning / Docker / tracking

_____

_____

7. Planned AI-Tool Usage

Mention the AI tools / models used for coding assistance and specify the purpose and project artifact affected. This information will be carried into the final AI-tool usage declaration.

AI Tool / Model

_____

Purpose

_____

Artifact Affected

_____

Student Declaration

☐ I will not change the approved case-study scope, major methodology, or implementation approach without prior faculty approval.

☐ I confirm that the work submitted under this synopsis represents my own understanding and contribution, and I will be responsible for explaining every significant part of the selected case study and its implementation.

Name and Signature: _____ Date: _____